



ETE 2721 L: Guitar Amplifier PCB Project
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INTRODUCTION:

The goal of this project was to build a Printed Circuit Board by creating a design of our own with Altium Designer. After doing so, we would transfer that design into Fusion 360 and create a 3D enclosure to protect our design. By accomplishing this goal, we would gain a better understanding of how a PCB is created on CAD software and how the design and manufacturing process work. Our project was building a Guitar Amplifier that would allow an electroacoustic guitar to output sound through a speaker, with adjustable volume and gain.

CIRCUIT OPERATION:

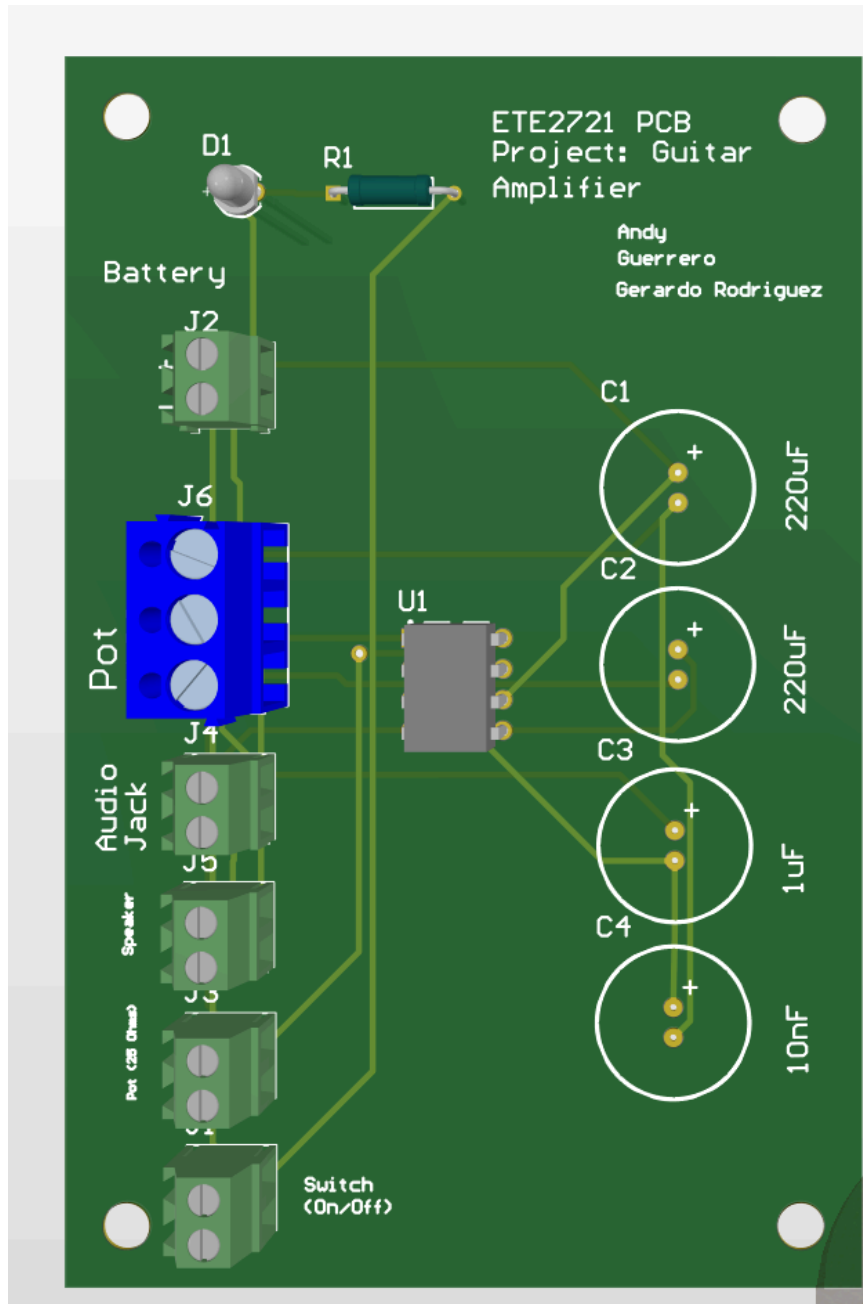
Our circuit, the Guitar Amplifier, has an input audio jack that would have a signal that is amplified through an LM386 op amp, and output sound through an 8-ohm speaker that has adjustable volume and adjustable gain. The way the circuit works is by having a 9V battery powering the circuit. It then has a switch that lets the user turn the power on or off. The LED lets the user know if the circuit is powered on and provides reverse polarity protection, and the resistor protects the current going into the diode. After all of this, there is a decoupling capacitor in place to reduce power supply noise. There is then the audio source that connects to the circuit (which in this case, is a guitar) via an audio jack. The signal then passes through a capacitor that blocks any DC component that might interfere with the signal amplification. The signal passes through a 25 Ω potentiometer which acts as the gain control of the signal before it enters the LM386. The signal enters the op amp through pin 3 (non inverting input). Pin 2 is then grounded (which is the inverting input). The amplifier's gain is then further adjusted by another capacitor which is then connected between pins 1 and 8. With this capacitor, the gain is significantly higher, allowing for stronger amplification of small input signals. The LM386 then amplifies the input audio and outputs it via pin 5. Another capacitor is then placed in series with the output to block any unwanted parasitics, and allow the signal to pass on through to the speaker, which is also connected to a 100 k Ω potentiometer that controls the volume.

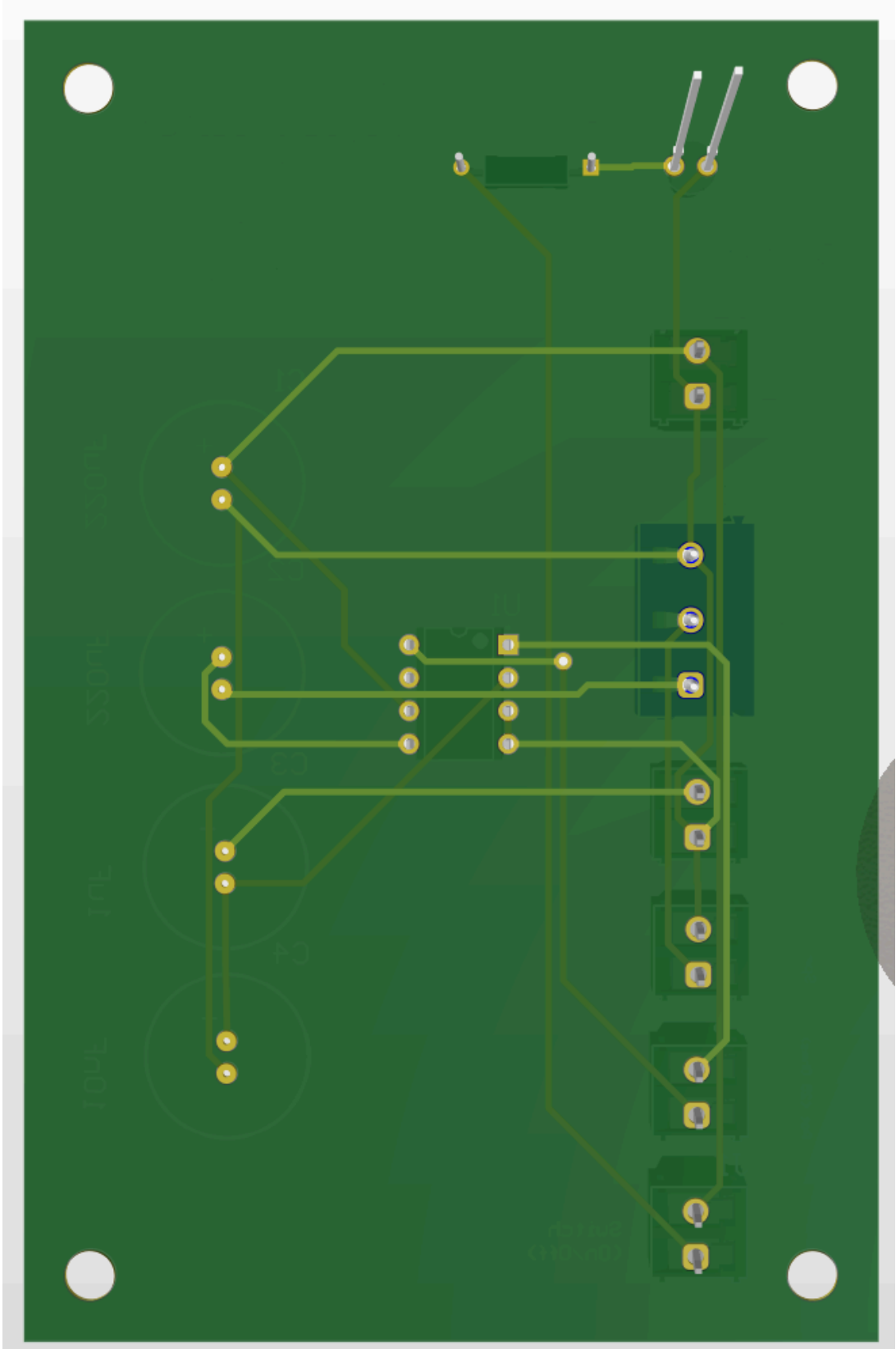
The diagram illustrates a circuit for an audio amplifier. It features an LM386N-4 op-amp (U1) configured as a voltage follower. The non-inverting input (+) is connected to the wiper of a 100K potentiometer (J6) and to a 10k resistor (R1) connected to a 9V battery (J2). The inverting input (-) is connected to the wiper of a 25 Ohms potentiometer (J3) and to the output of the op-amp. The output of the op-amp is connected to a speaker (J5) and to a 100k potentiometer (J6). The circuit is powered by a 9V battery (J2) and includes several capacitors (C1, C2, C3, C4) for filtering and coupling. A switch (J1) is used to turn the circuit on and off. The components are labeled with their respective values and part numbers.

PCB Creation Process:

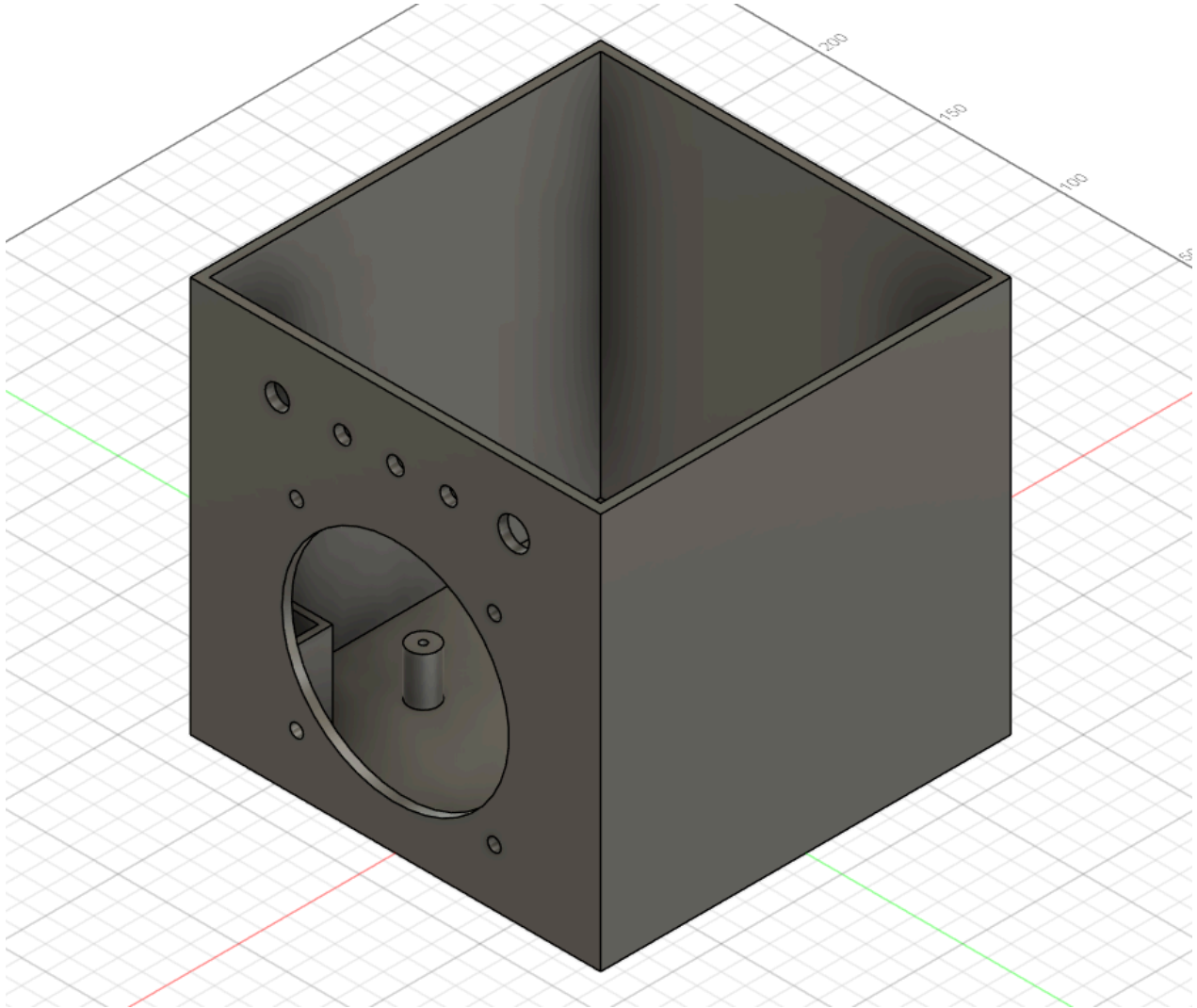
In the PCB creation process, we had a prototype already made beforehand. It already worked, but it was a bit of a squirrel's nest and we realized from there that making a PCB for it would make the project significantly cleaner. A lot of the project parts were already bought, as we already had spare parts laying around from my previous version, so that was another benefit of this project. After deciding on this project and commenced the designing process, we realized how much of a hassle it was to download and find footprints for the components we were going to use for our project. We had to think three dimensional, and we also asked Professor Mandala what we should do if we are struggling to find components on the Manufacturer Part Search. He recommended I just find components with similar sizes ready to go. After realizing this, we went ahead and found those components and nearly completed the circuit. We realized that for the speaker, potentiometers, and audio jack, we would need terminal blocks that would allow me to connect the wires easier and adjust the reach to where I needed them to be in our box for our guitar amp. We also noticed a design flaw with the help of Professor Mandala that I also had placed my LED in the wrong direction. After designing the schematic, we went ahead and finished designing the PCB by using Autorouting. We double checked that every wire connected to where they should be and tried to make everything as compact as possible. After doing this, we went ahead and put our names on our board with silkscreen and put the name of our class. After doing so, we ordered the PCB on JLCPCB. After ordering, we realized that we should have added the CPP logo and had each name on one line. Additionally, we should have modified the holes for the PCB, but when we received the PCB, it turned out alright. Everything looked the way we wanted it, albeit some minor changes we should have made.

PRINTED CIRCUIT BOARD DESIGN:

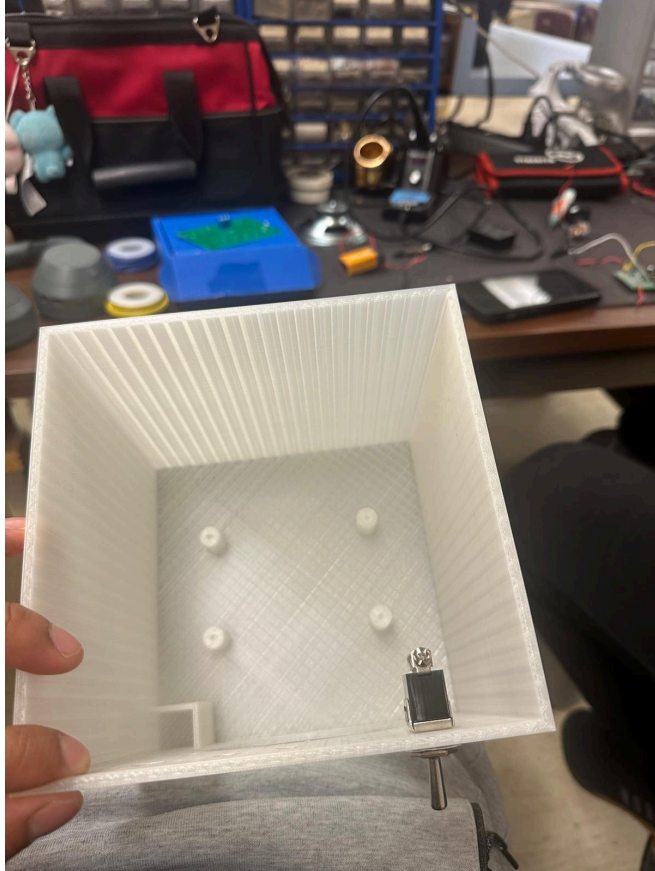




3D MODEL ENCLOSURE DESIGN:



PICTURES OF FINAL DESIGN:





DESIGN CONSIDERATIONS:

Several important considerations were taken into account during the design of the guitar amplifier PCB. First, we prioritized keeping the board layout compact and efficient to fit comfortably inside a custom 3D-printed enclosure. We also needed to ensure that all components had available footprints in Altium Designer, which meant substituting some parts for similarly sized alternatives when exact matches were not available.

Thermal management was considered for components like the LM386, which may generate heat during operation. Proper spacing and placement of the decoupling capacitors were also crucial to reduce noise and improve overall stability. We placed special attention on the routing of signal paths to minimize interference, especially for the input and output audio lines.

A key part of the success of our design was the strategic placement of the PCB at the center of the enclosure. This positioning, combined with the use of screw terminals for major connections like the speaker, potentiometers, and audio jack, allowed us to connect these components using wires and position them freely wherever needed inside the enclosure. This flexibility greatly improved cable management and made the final assembly much more adaptable and organized.

BOM:

Comment	Description	Designator	Footprint	LibRef	Quantity	Cost
ESC227M010AE3AA	Aluminum Electrolytic Capacitor, 227µF, 10%, 50V	C1, C2, C3	ESC227M010AE3AA-Footprint	CMP-001-00084-1	4	\$8.00 (kit)
151031VS06000	Single Color LED, Brightness 15000mcd	D1	WL-TMRC_D3.95H5.3	CMP-007-00044-1	1	(on hand)
691214110002	Barrier Strip Terminal Block, 14 Positions	J1, J2, J3, J4, J5	691214110002	CMP-002-00074-1	5	\$7.68
691101710003	Barrier Strip Terminal Block, 7 Positions	J6	691101710003	CMP-002-00072-1	1	(on hand)
MRS25000C1001FCT00	Fixed Resistor, Thin Film, 25000Ω, 1%	R1	FP-MRS25-MFG	CMP-009-00186-1	1	(on hand)
LM386N-4	Audio Amplifier, 1W, 14 Pin DIP	U1	TI-P8	CMP-014-00005-1	1	(on hand)
3D Print	PVA	N/A		N/A	1	\$35.00
					Total:	\$50.68

Our BOM that we got from Altium was not that costly. A lot of the parts we already had on hand, so the most expensive part of our project was the 3D print for the enclosure.

CONCLUSION:

This project gave us hands-on experience with the full design process of a PCB, from schematic creation to physical assembly. We successfully designed, simulated, and manufactured a working guitar amplifier that functions as intended, with adjustable volume and gain controls. Throughout the process, we learned how to troubleshoot design flaws, adapt to available components, and optimize our PCB layout for both electrical performance and mechanical fit.

Creating the 3D model for the enclosure further deepened our understanding of how electronics and mechanical design must work together. The final product met our expectations. Overall, this project was an excellent opportunity to apply our skills in PCB design, CAD modeling, and practical electronics in a meaningful and rewarding way.